

EXP-3Demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.

```
import pandas as pd
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.tree import DecisionTreeClassifier, export_text
```

```
# Play Tennis Dataset
```

```
data = {
```

```
    'Outlook':
```

```
['Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast','Sunny','Sunny','Rain','Sunny','Overcast','Overcast','Rain'],
```

```
    'Temperature':
```

```
['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild','Cool','Mild','Mild','Mild','Hot','Mild'],
```

```
    'Humidity':
```

```
['High','High','High','High','Normal','Normal','Normal','High','Normal','Normal','Normal','Normal','High','Normal','High'],
```

```
    'Wind':
```

```
['Weak','Strong','Weak','Weak','Weak','Strong','Strong','Weak','Weak','Weak','Strong','Strong','Weak','Strong'],
```

```
    'PlayTennis': ['No','No','Yes','Yes','Yes','No','Yes','No','Yes','Yes','Yes','Yes','Yes','No']
```

```
}
```

```
df = pd.DataFrame(data)
```

```
print("Training Dataset:\n")
```

```
print(df)
```

```
# Encode categorical data
```

```
le = LabelEncoder()
```

```
encoded_df = df.copy()
```

```
for column in encoded_df.columns:
```

```
    encoded_df[column] = le.fit_transform(encoded_df[column])
```

```
X = encoded_df.iloc[:, :-1]
```

```
y = encoded_df.iloc[:, -1]
```

```
# Train Decision Tree using ID3 (Entropy)
```

```
clf = DecisionTreeClassifier(criterion='entropy')
```

```
clf.fit(X, y)
```

```

# Display Decision Tree

print("\nDecision Tree Rules:\n")

print(export_text(clf, feature_names=list(X.columns)))

# New Sample

new_sample = pd.DataFrame({

    'Outlook':['Sunny'],

    'Temperature':['Cool'],

    'Humidity':['High'],

    'Wind':['Strong']})

# Encode new sample

encoded_sample = new_sample.copy()

for col in new_sample.columns:

    le = LabelEncoder()

    le.fit(df[col])

    encoded_sample[col] = le.transform(new_sample[col])

prediction = clf.predict(encoded_sample)

print("\nNew Sample:")

print(new_sample)

if prediction[0] == 1:

    print("\nPrediction: Play Tennis = Yes")

else:

    print("\nPrediction: Play Tennis = No")

```

Training Dataset:

```

---
   Outlook Temperature Humidity Wind PlayTennis
0    Sunny         Hot      High  Weak         No
1    Sunny         Hot      High Strong         No
2  Overcast         Hot      High  Weak         Yes
3     Rain      Mild      High  Weak         Yes
4     Rain      Cool    Normal  Weak         Yes
5     Rain      Cool    Normal Strong         No
6  Overcast      Cool    Normal Strong         Yes
7    Sunny      Mild      High  Weak         No
8    Sunny      Cool    Normal  Weak         Yes
9     Rain      Mild    Normal  Weak         Yes
10   Sunny      Mild    Normal Strong         Yes
11  Overcast      Mild      High Strong         Yes
12  Overcast         Hot    Normal  Weak         Yes
13     Rain      Mild      High Strong         No

```

DECISION TREE RULES:

```

---
--- Outlook <= 0.50
|   |--- class: 1
|   |--- Outlook > 0.50
|   |   |--- Humidity <= 0.50
|   |   |   |--- Outlook <= 1.50
|   |   |   |   |--- Wind <= 0.50
|   |   |   |   |   |--- class: 0
|   |   |   |   |   |--- Wind > 0.50
|   |   |   |   |   |   |--- class: 1
|   |   |   |   |--- Outlook > 1.50
|   |   |   |   |   |--- class: 0
|   |   |   |--- Humidity > 0.50
|   |   |   |   |--- Wind <= 0.50
|   |   |   |   |   |--- Outlook <= 1.50
|   |   |   |   |   |   |--- class: 0
|   |   |   |   |   |   |--- Outlook > 1.50
|   |   |   |   |   |   |   |--- class: 1
|   |   |   |   |   |--- Wind > 0.50
|   |   |   |   |   |   |--- class: 1
|   |--- class: 0
|--- class: 1

New Sample:
Outlook Temperature Humidity Wind
0 sunny         Cool      High  Strong

Prediction: Play Tennis = No

```